Higher Math HW3

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October 19, 2026

1. Consider a simplified fragment of the genetic code. Let

$$C = \{\text{AAA}, \text{AAG}, \text{UUU}, \text{UUC}, \text{GGG}\}, \quad P = \{\text{Lys}, \text{Phe}, \text{Gly}, \text{Trp}\}.$$

Define $f : C \to P$ using the table:

Codon	AAA	AAG	UUU	UUC	GGG
f(Codon)	Lys	Lys	Phe	Phe	Gly

(a) Determine the image $f(C)$.
$im(f(C)) = \{f(x) : x \in C\} = \{Lys, Phe, Gly\}$

(b) Is f injective? Explain.
f is injective if $f(x_1) = f(x_2) \iff x_1 = x_2$. However $f(AAA) = f(AAG) = Lys$. Thus, f is not injective

(c) Is f surjective onto P? Explain.
f is not surjective as $Trp \in P$, but there is not such $c \in C$ $that f(c) = Trp$

(d) Find $f^{-1}(\{\text{Lys}\})$
$f^{-1}(\{Lys\}) = \{AAA, AAG\}$

(d) Would $f : C \to f(C)$ be surjective? What changed?
Since $C = \{AAA, AAG, UUU, UUC, GGG\}$ and $f(C) = \{Lys, Phe, Gly\}$, mapping $f : C \Rightarrow f(C)$ is surjective as for $\forall y \in f(C) \exists x \in C | f(x) = y$

2. Let $f, g : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x - 1$ and $g(x) = x^2$.

(a) Find $(g \circ f)(x)$.
$(g \circ f)(x) = g(f(x)) = (2x - 1)^2$

(b) Find $(f \circ g)(x)$.
$(f \circ g)(x) = f(g(x)) = 2x^2 - 1$

(c) Evaluate both compositions at $x = 2$.
$(g \circ f)(2) = 3^2 = 9$
$(f \circ g)(2) = 8 - 1 = 7$

(d) Does $f \circ g = g \circ f$ hold? Explain.

Assume $f \circ g = g \circ f$. Then $2x^2 = 1 = (2x-1)^2$

$x^2 - 2x + 1 = 0$

$(x-1)^2 = 0$

Thus, $f \circ g = g \circ f$ holds iff $x = 1$. Thus it doesn't hold in general case

3. A laboratory assigns samples to sequencing batches and batches to quality-control (QC) statuses. Let $S = \{s_1, s_2, s_3, s_4\}$, $B = \{b_1, b_2, b_3\}$ and $Q = \{\text{pass}, \text{review}\}$. Define $f : S \to B$ and $g : B \to Q$ by

$$f(s_1) = b_1, \quad f(s_2) = b_1, \quad f(s_3) = b_2, \quad f(s_4) = b_3,$$
$$g(b_1) = \text{pass}, \quad g(b_2) = \text{review}, \quad g(b_3) = \text{pass}.$$

   (a) Construct the mapping $g \circ f$ (give its values at all samples).

   $g(s) = g(f(s)) : S \Rightarrow Q$ $g(f(s_1)) = pass$ $g(f(s_2)) = pass$ $g(f(s_3)) = review$ $g(f(s_4)) = pass$

   (b) State its domain and codomain.

   $D(g(f(s)) = \{s_1, s_2, s_3, s_4\}$

$CD(g(f(s)) = \{pass, review\}$

(c) Is $g \circ f$ injective? Is it surjective? Explain.

$(g \circ f)(s)$ is not injective, as $g(f(s_1)) = g(f(s_2)) = pass, s_1 \neq s_2$ $(g \circ f)(s)$ is surjective, as $\forall y \in CD(g(f(x)) \exists x \in D(g(f(x)) | g(f(x)) = y$

(d) Explain why $f \circ g$ is not defined for these mappings.

Assume mapping $f \circ g$ is defined. Then codomain Q of g must be a subset of domain S of f. However, $Q \not\subseteq S$. Contradiction found. Thus, mapping $f \circ g$ is not defined

4. Work with the following mappings.

   (a) For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x-2$, find $f^{-1}(x)$ and verify both identities $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$.

   $f^{-1}(x) = \frac{x+2}{3}$

   $f^{-1}(f(x)) = \frac{3x-2+2}{3} = x$

   $f(f^{-1}(x)) = 3(\frac{x+2}{3}) - 2 = x + 2 - 2 = x$

   (b) Explain why $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, has no inverse mapping.

   Consider $x_1 = 2, x_2 = -2$. Thus $f(-2) = f(2) = 4, x_1 \neq x_2$. Thus, $f(x) = x^2$ is not injective. Thus, it is not bijective. Thus, its inverse mapping is undefined

   (c) Consider instead $f : [0, +\infty) \to [0, +\infty)$, $f(x) = x^2$. Does an inverse exist? Find it if possible.

   Assume $f(x_1) = f(x_2)$. Then $x_1^2 = x_2^2$. Since $D(f(x)) = [0, +\infty), x_1 = x_2$. Thus, mapping is injective.

   Mapping $f(x)$ is also surjective as $\forall f(x) \in [0, +\infty) \exists x \in [0, +\infty) | x^2 = f(x)$

Since mapping is surjective and injective, its bijective. Thus, its reverse function exists

$f^{-1}(x) = \sqrt{x}$

$f^{-1}(f(x)) = \sqrt{x^2} = x$

$f(f^{-1}(x)) = (\sqrt{x})^2 = x$

(d) Find the inverse of the mapping $1 \mapsto 3, 2 \mapsto 1, 3 \mapsto 4, 4 \mapsto 2$ from $\{1, 2, 3, 4\}$ to itself.

The inverse mapping of such mapping is defined as follows:

$1 \Rightarrow 2$

$2 \Rightarrow 4$

$3 \Rightarrow 1$

$4 \Rightarrow 3$

5. Let $f : A \to B$ and $g : B \to C$.

(a) Prove that if f and g are injective, then $g \circ f$ is injective.

Assume $g \circ f$ is not injective

Thus, $g(f(x_1)) = g(f(x_2)); x_1 \neq x2$

Since g in injective, $g(f(x_1)) = g(f(x_2)), f(x_1) = f(x_2)$

Therefore, $f(x_1) = f(x_2)$

Since f is injective, $f(x_1) = f(x_2), x_1 = x_2 \Rightarrow$ *contradiction.*

Thus if f and g are injective, $g \circ f$ is injective

(b) Prove that if f and g are surjective, then $g \circ f$ is surjective.

Suppose $z \in C$. Sinse g is surjective, $\forall z \in C \exists y \in B | g(y) = z$. Since f is surjective, $\forall y \in B \exists x \in A | f(x) = y$. Combined, $\forall z \in C \exists x \in A | g(f(x)) = z$. Thus, is g and f are surjective, $g \circ f$ is surjective